

MorphCubes: Adaptive Modular Robots for Dynamic Remote Human Embodiment

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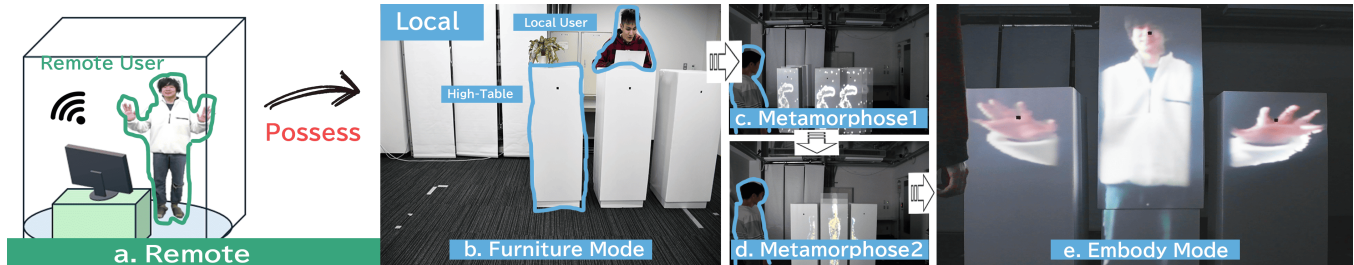


Figure 1: MorphCubes overview. (a) A remote user possessing the furniture in the local place. (b) A local user working on the modular furniture (e.g., counter). (c&d) Metamorphosing the modular furniture into modular robots with dynamic projection mapping. (e) The metamorphosed three modular robots represent the remote user's body and arms through their movements and height-changing.

Abstract

This study proposes MorphCubes, a robotic telepresence system that metamorphoses modular furniture arranged in the local environment into coordinated modular robots, enabling the physical representation of a human-scale embodiment of a remote user. Each robotic module replicates the remote user's body parts and movements, such as arm gestures and head rotations, through its physical locomotions and adapted dynamic projection mappings. This facilitates richer non-verbal communication involving a full-body physical presence as if the remote user physically presents locally. Leveraging the modularity of the robots, the system offers sufficient flexibility in the body expressions and cost-efficiency as they revert to their original furniture rolls after the telepresence session ends. In this late-breaking work, we discuss the design and implementation of our proof-concept prototype system with three modular robots and show its potential and flexibility through representative application scenarios, including hybrid meetings, remote tactile communication, and collaborative work.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Displays and imagers*; Ubiquitous computing.

Keywords

Dynamic projection mapping, shape-shifting robotic furniture

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1 Introduction

Conventional remote communication systems relying on video and audio transmissions allow accurate representation of a remote user's face and voice. However, non-verbal elements such as body orientation, eye direction, and hand gestures are often lost, making it difficult to perceive the presence of the remote user and foster a shared sense of space.

We propose MorphCubes, a novel telepresence system that utilizes multiple modular robots embedded in the environment to convey the physical presence of remote users. By integrating mobility, transformation capabilities, and dynamic projection mapping, MorphCubes metamorphoses modular furniture—such as partitions, tables, and counters commonly found in homes and offices—into robotic modules (Figure 1(b)). These robotic modules enable life-sized, physical representations of the remote user's body and hand movements through coordinated robots' motions and dynamic projection mapping. This creates an impression, much like a hologram, that the remote user has possessed the local furniture (Figure 1(e)), facilitating natural spatial arrangements for effective face-to-face

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interactions. Importantly, the system repurposes furniture temporarily so all robots can return to their original function once the telecommunication concludes.

While robotic furniture is not yet ubiquitous, devices like robotic vacuum cleaners and automated serving carts are becoming increasingly common in homes, offices, and restaurants. Given the active research efforts exploring robotic furniture capable of movement and shape-changing [4, 15, 21, 22, 28, 30], we anticipate that near-future workspaces may increasingly incorporate robotic elements. Indeed, early examples of robotic furniture have already been proposed to support virtual workspace scenarios [29, 33].

In this work, we design and implement a proof-of-concept prototype of MorphCubes that features three modular robots, dynamic projection mapping, wireless data communication, and various sensing platforms. To illustrate the system’s potential, we present representative application scenarios, including remote social interaction, hybrid meetings, and social touch. These examples would be discussion points in technical and practical implications for future robotic telepresence systems.

2 Related Work

Many researchers have explored robotic-based telepresence systems to enhance the sense of presence for remote communication. A notable example is humanoid communication robots, which convey not only voice and facial information but also detailed physical movements of the arms and fingers by activating their joint actuators (e.g., the Telesar series[32], the CB-i humanoid robot[13]). There is another approach to enhance the presence of remote users using a humanoid robot that closely resembles the remote user [10]. While these methods offer substantial performances, they remain challenging due to the high preparation costs for the robots.

Another approach involves robots that physically express only the essential elements needed for effective interaction. The face is crucial for expressing emotions, while the body—particularly its height—plays a significant role in defining physical characteristics (e.g., physical characteristics, age, etc.) and establishing a sense of presence within a given space [6, 24]. By utilizing a vertically adjustable camera mounted on a linear actuator, it becomes possible to represent the remote user’s height and adjust their gaze dynamically, thereby enhancing communication [19]. Similarly, the arms are vital not only for gestures but also for emotional expression[20]. A unique robotic table has been proposed that physically replicates the remote user’s hand and arm movements using multiple up-down actuators, enabling tactile feedback and object manipulation in shared workspaces[16]. Furthermore, SAR (Spatial Augmented Reality)-based telepresence systems, which represent life-sized remote users, can create the illusion of co-presence, enabling natural face-to-face communication through auditory and visual engagement[7, 11, 14, 23]. HoloBots synchronizes holograms displayed on AR glasses with the movements of on-desk robots, seamlessly integrating visual and physical actions within the same space [9].

The idea that remote users possess robotic cameras has been proposed. For example, some systems utilize robotic or human-mounted sensors or cameras in remote physical environments,

allowing remote users to selectively use those cameras and get arbitrary viewpoints for remote space explorations[12, 18].

Swarm robots exhibit high robustness and flexibility, enabling the creation of diverse interfaces through actions such as aggregation and separation[5, 31]. Recently, studies have focused on enhancing the sense of presence using swarm robots. Swarm Body[8] uses a swarm of small robots to physically replicate the remote user’s hand movements, allowing for tasks like rock-paper-scissors or object transportation in physical spaces. These systems, which involve coordinated motions of the modular robots, offer not only visual representation but also tactile feedback. Notably, this research has effectively replicated human-like motions and a sense of presence through non-human-shaped robots. However, existing proposals often rely on small robots that do not convey life-sized physical presence.

3 Design of MorphCubes

3.1 Concept and goal

Figure 1 illustrates the concept of MorphCubes with three modular robot instances. This research aims to simplify the physical embodiment of remote users (Figure 1(a)) by metamorphosing furniture into modular robots. By repurposing such life-size furniture as tables, chairs, and shelves, often arranged in modular patterns, the system enables human-scale representation of the remote user’s body movements.

In our system, a remote user can “posses” a set of modular furniture at a local location (see Figure 1(b)). Once connected, this furniture transforms into modular robots (refer to Figures 1(c-d)). These robotic modules synchronize their movements with the body and hand motions of the remote user, physically recreating the user’s presence in the local space (illustrated in Figure 1(e)). The system captures key attributes of the remote user, such as height and spatial arrangements, which influence communication. Additionally, it utilizes the characteristics of swarm robotics to enhance hand and body movements. For instance, the representation of the remote user’s hands can extend beyond their physical structure, allowing for the presentation of information or tactile interactions with users at a distance. After the interaction is completed, the modular robots seamlessly return to their original positions and resume their function as furniture.

As discussed, our work builds upon extensive prior research in telecommunication; however, our primary contribution lies in the novel concept of transforming modular furniture into remote human surrogates. Leveraging furniture’s inherent physical presence in the space, this approach can express richer and more realistic human presence and haptic interactions through prominent social and non-verbal movements, compared to projection-based or virtual avatar systems.

3.2 Design considerations

3.2.1 Modularity. The system should be flexible regarding body representation and the roles of furniture. We utilized primitive cube-shaped modular robots as our examples. When the robotic functions are disabled, these robots can serve as a high table, a counter, or shelves in office spaces.

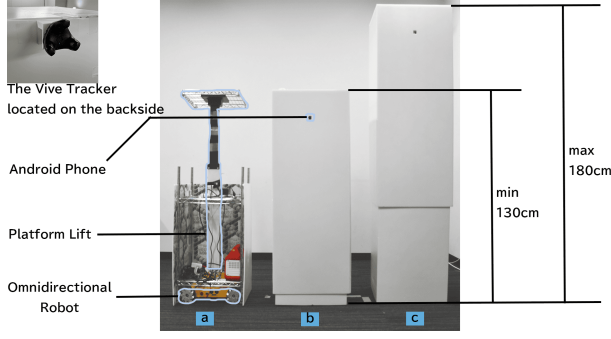


Figure 2: Modular robot design

The modular robots are designed with an adjustable height ranging from 130 cm to 180 cm, with a width and depth of 50 cm each. The two robots shown in Figure 2(b-c) illustrate the minimum and maximum height configurations, respectively. The minimum height version resembles a high table at 130 cm, suitable for a workspace or home setting. However, when a remote user “processes” the system, the modular robots adapt to represent the user’s body features—such as height, face position, and both hands—at full scale within the 180 cm height range.

3.2.2 Body part representation. We used three modular robots to achieve a minimal body representation. While increasing the number of modular robots can provide a more detailed representation—such as adding more joints—this can also complicate the coordination among the robots. Furthermore, in research related to embodiment and the sense of agency in avatars, it remains unclear what the optimal number of body parts is for effective communication. Therefore, we focus first on the three body parts that are essential for conveying emotions, non-verbal cues, and bodily movements.

3.2.3 Visual presentation. Due to the limited expression capabilities of the metamorphosed modular robot’s physicality, We introduce additional visual overlay technologies that represent the remote user’s appearance, enabling local users to better understand their impression, facial features, style, and simple emotional expressions. The idea is behind our expectation that various SAR technologies (e.g., AR glass, projection mapping) will become more ubiquitous[3].

Additionally, to clearly convey the functions of each body part, this study employs a method that utilizes projectors installed in the environment to project visuals. While AR glasses are an option, their narrow field of view (FOV) does not suit our needs. The system continuously manages the movements and positions of the robots, and based on this information, dynamic projection mapping is applied to the front surfaces of the modular robots, resulting in a realistic physical representation.

3.2.4 Safety and unobtrusive operation. Our life-sized modular robots are designed to focus on safety without causing collisions with users and other modules. Due to their heavy mass and high kinetic energy, precautions have been taken to prevent accidents and to foster a sense of reassurance during use. One consideration

is to avoid surprising or alarming users when the furniture suddenly metamorphoses into modular robots and starts moving. For this, we carefully designed the projection content that involves the transition animations to clearly depict the metamorphosis progress from “furniture mode” to “embody mode”. It makes an unobtrusive interaction with the local users (Figure 1(c)).

4 MorphCubes Prototyping

4.1 Hardware

Figure 2(a) illustrates the robot’s mechanism. The weight of each modular robot is 26 kg. We adopted the Nexus Robot (4kg) to realize omnidirectional movement. The ESP32 was used for wireless communication with the motors, while an Arduino Atmega328 inside the Nexus Robot communicates with the ESP32 to control the four wheels’ motors. Another ESP32 wirelessly controls the electric lifting mechanism used to control the height of the module. We employed a Lighthouse tracking system to monitor the modular robot’s position. When the hand moved sideways at a speed of 30 cm/s, we measured the speed and delay of the modular robot. The robot achieved a speed of 20 cm/s, and the average delay between the remote user’s hand movement and the robot was 0.57 seconds during the initial movement. This delay increased to 1.72 seconds when the hand came to a stop, indicating a slight rise compared to the initial delay. While high-frequency gestures, such as rapid hand waving, remain challenging for the system to replicate, it effectively transmits non-repetitive gestures like high-fives and hugs. Therefore, it is considered adequate for conveying nonverbal communication. The modular robot features a 2 cm × 2 cm square opening on its front panel, as shown in Figure 2(b). The views from these cameras are transmitted to a remote system. For dynamic projection mapping, a BenQ TK700STi 4K short-throw projector was utilized, positioned to ensure that the projection area covers the entire operational range of the three modular robots.

Figure 3(a) illustrates the remote setup. The body movements of the remote user are captured using the Kinect v2, which consists of a depth camera and an RGB camera. This data is then structured to provide skeletal information. The user’s body gravity, including movements such as tilting forward, backward, and side to side, is monitored using the Wii Balance Board. The sensory data is transmitted to a local location and utilized to control the body robot. The full-body platform presented in this implementation serves as an example and can be modified further. However, one important requirement is that remote users should retain the freedom to use their hands and fingers for intuitive gestures [27].

Figure 3(b) illustrates the local setup in which three modular robots are operated, along with a projector and a 3D tracking system. We have also installed a ceiling camera to provide the remote user with essential spatial awareness of the local environment and the relationships among the modular robots. Figure 3(b) shows example views of the on-robot Android and ceiling cameras when using hybrid meeting scenarios. These camera feeds, along with audio, are transmitted to the remote system via Zoom, a widely used video conferencing and streaming platform.

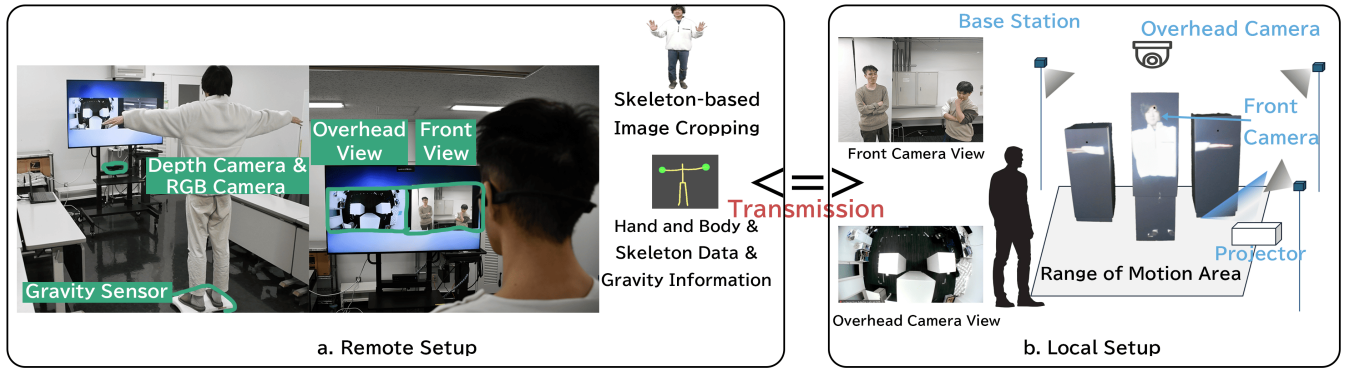


Figure 3: Remote and local setups

4.2 Software and workflow

4.2.1 Robot control method. The Unity software controls the modular robots by transmitting the remote user's skeletal data and the gravity via UDP communication (Figure 3(b)). The modular robots control in the two modes:

- (1) **Furniture Mode:** This mode is used during the initial setup of the modular robots or when repositioning them after ending the embodiment representation. The PID control is employed to accurately align the modular robots in parallel to predefined positions in Unity. The RVO (Reciprocal Velocity Obstacles) algorithm is adopted to simulate real-time movement directions to prevent collisions between multiple modular robots.
- (2) **Embody Mode:** The robotic module's forward, backward, and lateral movements depend on the remote user's gravity. When the gravity tilts beyond a certain threshold, the velocity of the body robot is calculated accordingly and transmitted to the modular robots. The body robotic module shares a single coordinate system with both hand robotic modules. The positions of both hands, obtained from the remote user's skeletal information, are applied to the hand robot's controls in real-time. These modules follow the most recent hand positional data; however, they do not rigidly mimic the movements of the hands. Instead, they calculate efficient paths and take shortcuts to optimize motion. Additionally, the yaw rotation of the head, based on the skeletal data, determines the orientation of the body robot.

4.2.2 Dynamic Projection Mapping. We used TouchDesigner, a node-based visual programming environment for dynamic projection mapping onto the robotic modules. Dynamic projection mapping requires the optical axis of the projector in the physical space to precisely align with the optical axis of the virtual camera in the virtual space. In this study, we achieved fast and precise calibration using TouchDesigner's built-in "camShnappr," which estimates the camera position in the virtual space by reverse-calculating from the shape data of the projection target. The communication between Unity and TouchDesigner is performed via OSC (Open Sound Control), and the modular robot's position is synchronized with the projector's visuals by sending positional data from the Vive Tracker to TouchDesigner.

The content projected by the projector is processed in real-time using body and hand images, which are extracted based on skeletal information using Kinect, and this information is transmitted to the local system via WebRTC.

5 Application Scenarios

We propose a set of application scenarios that leverage the unique features of MorphCubes. These scenarios demonstrate how our MorphCubes can be used in the new telecommunication scenarios while maintaining spatial interaction aspects in the local spaces. Primarily, they focus on non-verbal communication, such as the distance to remote users, gestures, and haptics as social interactions with partners.

5.1 Social and tactile communication

Social touch makes it possible to bridge the psychological gap and to communicate familiarity and reassurance. For example, hand robotic modules can gently touch a local person's shoulder or arm to indicate the intention to speak or provide reassurance. Similarly, these robots can perform social gestures such as high-fives and hugs, developing positive relationships in remote communication. Such touch gestures are expected to convey messages of celebration or encouragement within a team.

5.2 Explicit group formation for hybrid meeting

Figure 4(b) illustrates the dynamics of a hybrid meeting involving one remote and two local users. Group formation, involving body directions and each other, is a critical element that significantly affects the types and activities in the multi-user discussions[1]. This system physically expresses distance to remote users and directs the eye's direction by the physical positions of the modular robots. For instance, the body robots can approach the local users or turn in another direction, effectively conveying necessary attention among the teammates.

5.3 Physical interactions to local objects/partners

This system can enhance collaboration tasks during hybrid meetings by enabling the movement and organization of wheeled chairs and tables. Figure 4(c) shows a remote user handing over a chair

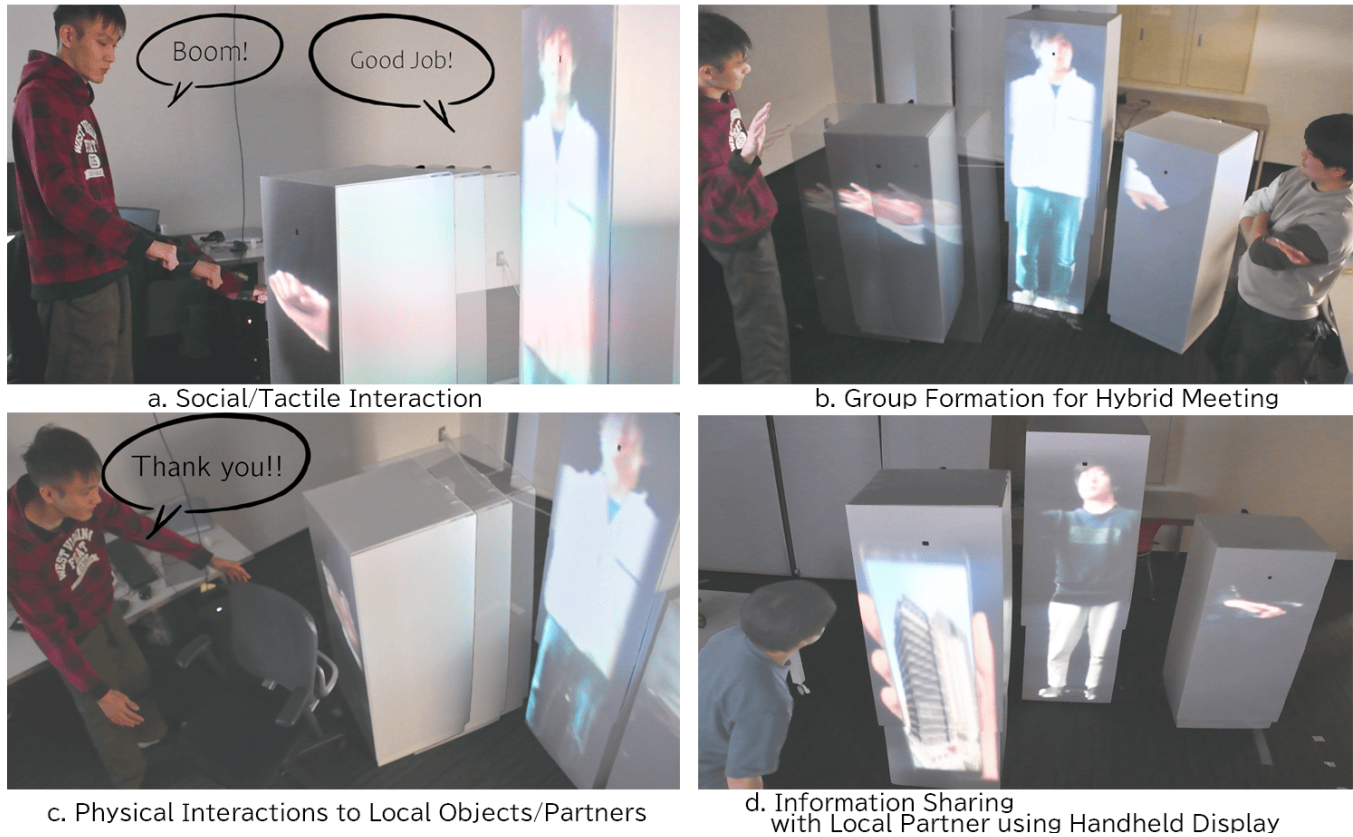


Figure 4: Application scenarios

with casters to a teammate who is present in the location. With its ability to interact physically with local furniture, this system can support collaborative spatial layout design among both local and remote users who share the same space and objects.

5.4 Hand-robot's display

This system can efficiently highlight points of interest by using hand robots as displays, presenting figures and documentation during online meetings (Figure 4(d)). By dynamically adjusting the height of their hands or extending their reach, remote users can enhance the impact of their messages, which might otherwise lack presence. This system also enables the selective presentation of information to specific users by adjusting the direction of the projected face. This feature presents new possibilities for communication that transcend the limitations of human physical structure, supporting smooth interactions while eliminating the need for mid-range physical movement.

6 Discussions

6.1 Technical limitations

During our iterative operations, we identified a significant issue: the delayed response of the modular robot's movements in relation to the actions of remote users. This delay is primarily attributed to the weight of the modular robot, which is 26 kg. Additionally,

challenges such as the transmission speed of full-body video and the slow elevation speed were noted.

These issues adversely affect the comfort of operation. One source of delay arises from the continuous real-time tracking of hand movements. To mitigate this delay, we could develop a remote user interface that selectively captures and translates specific portions of hand movements onto the modular robot.

The height adjustment range of this system, which goes up to 180 cm, is inadequate and poses a challenge. For comparison, the Scalable Body has successfully achieved a 100 cm extension using a reel-type actuator. Its lightweight design is also anticipated to help with weight-related issues. Moving forward, it's crucial to enhance the design of furniture robots by integrating lightweight mechanisms that allow for a wider height adjustment range.

6.2 Transition update

Our current safety system relies on unidirectional visual notifications from the robots to the users. Yet, some users may not notice them when they look at other places. Thus, it will be necessary to provide more apparent and substantial cues during transformations using additional modalities. Examples include the subtle motions of robots and sound, expressing that the robots are being transformed.

6.3 Transition Update

Our current safety system relies on unidirectional visual notifications from robots to users. However, users may fail to notice these when their attention is directed elsewhere. Therefore, it would be beneficial to provide more explicit cues through additional modalities during transitions. Examples include subtle robotic motions and auditory signals inexplicitly indicating that the robots are transforming.

6.4 User interface

In the early stages of prototype development, the remote user's viewpoint was limited to a front view only. However, a challenge arose as the exact position of the hand robot is unclear. This issue was thought to derive from the fact that the modular robot has only one camera view, and the field of view of that camera is narrow, which lacks a wide field of view like that of a human and a sense of body ownership. To address this, a "second eye" in the form of a ceiling camera was introduced.

An overhead view and a front view are insufficient for remote users to understand situational awareness and perceive the sense of ownership of the robotic modules. Our primary focus in this work was embodiment representation, but the next step could be an interface design that allows the user to freely position each robot. Achieving natural and intuitive operation was challenging for MorphCubes. Therefore, prioritizing a user interface that emphasizes operability through external cameras was deemed necessary.

These limitations highlight the need for innovative user interfaces. To address the issues of operability and delay, we propose a user interface that leverages input from the movement of the hand captured by the ceiling camera and overlays velocity lines as visual guidance. This approach can enable more intuitive and precise visual control of positions. The idea draws inspiration from interaction designs where a top-down ceiling view is displayed, allowing users to directly specify positions via their body gestures [17, 25].

Another viable approach involves using an HMD, which enables users to interact within a three-dimensional virtual environment, intuitively perceive their surroundings, and enhance their sense of body ownership. For instance, VRoxy proposes converting local physical spatial information into point-cloud data to construct a virtual environment that closely resembles the remote space, allowing users to physically walk around it [26]. Although the setup is somewhat more complex than the current system, it is feasible to incorporate HMD-based face tracking to convey facial expressions via virtual avatars.

6.5 User study plan

We plan to conduct a psychological user study to explore how people interpret human motions represented by modular robots, as well as their perceptions of current technical challenges (such as motion delay) and potential benefits (like full-body representation). We will consider various baselines for comparison. One intriguing baseline is metaverse applications that utilize avatars with appearances different from those of the actual users. Another valuable comparison point will be the display system itself. Augmented Reality (AR) or Virtual Reality (VR) can depict the avatars' bodies

and hand motions within the environment. In this context, key discussion points may include the limited field of view (FOV) of AR/VR glasses and the need for physical interaction or touch. To investigate the impact of furniture-type robots on users' sense of presence, a suitable approach would be to use the Networked Minds Measure of Social Presence [2]. This measure enables assessment of critical factors of social presence, such as co-presence and mutual attention.

7 Current Contribution Summary

The contribution of this late-breaking work is (1) proposing a novel concept of telecommunication system based on modular robotic furniture, which physically embody remote users through expressive movements., (2) showing the prototype's behaviors and functions through representative application scenarios, and (3) discussing future work focusing on the technical updates and social interaction user study.

References

- [1] Hrishav Bakul Barua, Theint Haythi Mg, Pradip Pramanick, and Chayan Sarkar. 2024. Enabling Social Robots to Perceive and Join Socially Interacting Groups using F-formation: A Comprehensive Overview. *ACM Transactions on Human-Robot Interaction* 13 (12 2024), Article 48, 1–48. Issue 4. <https://dl.acm.org/doi/10.1145/3682072>
- [2] Frank Biocca, Chad Harms, and Jenn Gregg. 2001. The Networked Minds Measure of Social Presence: Pilot Test of the Factor Structure and Concurrent Validity. *annual International Workshop on Presence* (1 2001), 1–9. https://www.researchgate.net/publication/200772411_The_Networked_Minds_Measure_of_Social_Presence_Pilot_Test_of_the_Factor_Structure_and_Concurrent_Validty
- [3] Andreas Rene Fender, Hrvoje Benko, and Andy Wilson. 2017. MeetAlive: Room-scale omni-directional display system for multi-user content and control sharing. *Proceedings of the ACM International Conference on Interactive Surfaces and Spaces* (10 2017), 106–115. doi:10.1145/3132272.3134117
- [4] Kazuyuki Fujita, Kazuki Takashima, Yuichi Itoh, and Yoshifumi Kitamura. 2023. Human-Workspace Interaction: prior research efforts and future challenges for supporting knowledge workers. *Quality and User Experience* 8 (2023), 1–19. Issue 1. doi:10.1007/s41233-023-00060-9
- [5] Mathieu Le Goc, Lawrence H. Kim, Ali Parsaei, Jean Daniel Fekete, Pierre Dragicevic, and Sean Follmer. 2016. Zooids: Building blocks for swarm user interfaces. *Proceedings of the Annual Symposium on User Interface Software and Technology* (10 2016), 97–109. <https://dl.acm.org/doi/10.1145/2984511.2984547>
- [6] Rita Gorawara-Bhat, Mary Ann Cook, and Greg A Sachs. 2007. Nonverbal communication in doctor-elderly patient transactions (NDEPT): development of a tool. *Patient education and counseling* 66 (5 2007), 223–234. Issue 2. doi:10.1016/j.pedc.2006.12.005
- [7] Keiji Hirata, Yasunori Harada, Toshihiro Takada, Shigemi Aoyagi, Yoshinari Shirai, Naomi Yamashita, Katsuhiko Kaji, Junji Yamato, and Kenji Nakazawa. 2008. T-Room: Next generation video communication system. *GLOBECOM - IEEE Global Telecommunications Conference* (2008), 5536–5539. doi:10.1109/GLOCOM.2008.ECP.1058
- [8] Sosuke Ichihashi, So Kuroki, Mai Nishimura, Kazumi Kasaura, Takefumi Hiraki, Kazutoshi Tanaka, and Shigeo Yoshida. 2024. Swarm Body: Embodied Swarm Robots. *Proceedings of Conference on Human Factors in Computing Systems* (2024), 1–19. doi:10.1145/3613904.3642870
- [9] Keiichi Ihara, Mehrad Faridan, Ayumi Ichikawa, Ikaku Kawaguchi, and Ryo Suzuki. 2023. HoloBots: Augmenting Holographic Telepresence with Mobile Robots for Tangible Remote Collaboration in Mixed Reality. *Proceedings of ACM Symposium on User Interface Software and Technology* (2023), 1–12. doi:10.1145/3586183.3606727
- [10] Hiroshi Ishiguro. 2023. Geminoid HI-6. <https://eng.irl.sys.es.osaka-u.ac.jp/projects/geminoid/>. 2025-01-16.
- [11] Brennan Jones, Yaying Zhang, Priscilla N.Y. Wong, and Sean Rintel. 2020. VROOM: Virtual robot overlay for online meetings. *Proceedings of Conference on Human Factors in Computing Systems* (4 2020), 1–10. <https://dl.acm.org/doi/10.1145/3334480.3382820>
- [12] Shunichi Kasahara and Jun Rekimoto. 2014. JackIn: Integrating first-person view with out-of-body vision generation for human-human augmentation. *Proceedings of the Augmented Human International Conference* (2014), Article 46, 1–8. doi:10.1145/2582051.2582097

- [13] Mitsuo Kawato. 2008. From 'Understanding the Brain by Creating the Brain' towards manipulative neuroscience. *Philosophical Transactions of the Royal Society B: Biological Sciences* 363 (6 2008), 2201–2214. Issue 1500. doi:10.1098/RSTB.2008.2272
- [14] Kibum Kim, John Bolton, Audrey Girouard, Jeremy Cooperstock, and Roel Vertegaal. 2012. TeleHuman: Effects of 3D perspective on gaze and pose estimation with a life-size cylindrical telepresence pod. *Proceedings of the Conference on Human Factors in Computing Systems* (2012), 2531–2540. <https://dl.acm.org/doi/10.1145/2207676.2208640>
- [15] Kent Larson, Hasier Larrea, Luis Alonso, and Carlos Rubio. 2015. CityOffice: What if your space was as dynamic as you are? <https://www.youtube.com/watch?v=3Z8rGjILX0>. 2025-03-05.
- [16] Daniel Leithinger, Sean Follmer, Alex Olwal, and Hiroshi Ishii. 2014. Physical Telepresence: Shape capture and display for embodied, computer-mediated remote collaboration. *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (2014), 461–470. doi:10.1145/2642918.2647377
- [17] Jiannan Li, Ravin Balakrishnan, and Tovi Grossman. 2020. StarHopper: A Touch Interface for Remote Object-Centric Drone Navigation. *Proceedings of Graphics Interface* (2020), 317–326. doi:10.20380/GI2020.32
- [18] Jiannan Li, Mauricio Sousa, Chu Li, Jessie Liu, Yan Chen, Ravin Balakrishnan, and Tovi Grossman. 2022. ASTEROIDS: Exploring Swarms of Mini-Telepresence Robots for Physical Skill Demonstration. *Proceedings of the Conference on Human Factors in Computing Systems* (4 2022), 1–14. doi:10.1145/3491102.3501927
- [19] Akira Matsuda, Takashi Miyaki, and Jun Rekimoto. 2017. ScalableBody: A telepresence robot that supports face position matching using a vertical actuator. In *Proceedings of the Augmented Human International Conference* (2017), Article 13, 1–9. doi:10.1145/3041164.3041182
- [20] Hiroyasu Miwa, azuko Itoh, Munemichi Matsumoto, Massimiliano Zecca, Hideaki Takariobu, Stefano Roccella, Maria Chiara Carrozza, Paolo Dario, and Atsuo Takanishi. 2004. Effective emotional expressions with emotion expression humanoid robot WE-4RII. *IEEE/RSJ International Conference on Intelligent Robots and Systems* (2004), 2203–2208. doi:10.1109/IROS.2004.1389736
- [21] Ken Nakagaki, Joanne Leong, Jordan L. Tappa, João Wilbert, and Hiroshi Ishii. 2020. HERMITS: Dynamically reconfiguring the interactivity of self-propelled TUIs with mechanical shell add-ons. *Proceedings of the Annual ACM Symposium on User Interface Software and Technology* (10 2020), 882–896. doi:10.1145/3379337.3415831
- [22] Yuki Onishi, Kazuki Takashima, Shoi Higashiyama, Kazuyuki Fujita, and Yoshifumi Kitamura. 2022. WaddleWalls: Room-scale Interactive Partitioning System using a Swarm of Robotic Partitions. *Proceedings of the Annual ACM Symposium on User Interface Software and Technology* (2022), Article 29, 1–15. doi:10.1145/3526113.3545615
- [23] Sergio Orts-Escolano, David Kim, Qin Cai, Christoph Rhemann, Philip Davidson, Philip Chou, Sean Fanello, Sameh Khamis, Sarah Mennicken, Wayne Chang, Ming-song Dou, Julien Valentin, Adarsh Kowdle, Vladimir Tankovich, Vivek Pradeep, Yury Degtyarev, Charles Loop, Shenlong Wang, Sing Bing Kang, Pushmeet Kohli, Yuliya Lutchyn, Cem Keskin, and Shahram Izadi. 2016. Holoportation: Virtual 3D teleportation in real-time. *Proceedings of the Annual Symposium on User Interface Software and Technology* (10 2016), 741–754. doi:10.1145/2984511.2984517
- [24] Irene Rae, Leila Takayama, and Bilge Mutlu. 2013. The Influence of Height in Robot-Mediated Communication. *Proceedings of the ACM/IEEE International Conference on Human-robot interaction* (2013), 1–8. doi:10.5555/2447556.2447558
- [25] Daisuke Sakamoto, Koichiro Honda, Masahiko Inami, and Takeo Igarashi. 2009. Sketch and run: A stroke-based interface for home robots. *Proceedings of the Conference on Human Factors in Computing Systems* (2009), 197–200. doi:10.1145/1518701.1518733
- [26] Mose Sakashita, Hyunju Kim, Brandon J. Woodard, Ruidong Zhang, and François Guimbretière. 2023. VRoxy: Enabling Remote Collaboration in Large Spaces Beyond Local Boundaries via a VR-Driven Robotic Proxy. *Proceedings of the Annual ACM Symposium on User Interface Software and Technology* (10 2023), 1–13. doi:10.1145/3586183.3606743
- [27] Mhd Yamen Saraiji, Charith Lasantha Fernando, Kouta Minamizawa, and Susumu Tachi. 2015. Mutual hand representation for teleexistence robots using projected virtual hands. *Proceedings of the Augmented Human International Conference* 11 (3 2015), 221–222. doi:10.1145/2735711.2735777
- [28] Alexander Spröwitz, Soha Pouya, Stéphane Bonardi, Jesse Van Den Kieboom, Rico Möckel, Aude Billard, Pierre Dillenbourg, and Auke Ijspeert. 2010. Roombots: Reconfigurable robots for adaptive furniture. *IEEE Computational Intelligence Magazine* 5 (8 2010), 20–32. Issue 3. doi:10.1109/MCI.2010.937320
- [29] Ryo Suzuki, Hooman Hedayati, Clement Zheng, James Bohn, Daniel Szafir, Ellen Yi-Luen Do, Mark D Gross, and Daniel Leithinger. 2020. RoomShift: Room-scale Dynamic Haptics for VR with Furniture-moving Swarm Robots. *Proceedings of the CHI Conference on Human Factors in Computing Systems* 20 (2020). doi:10.1145/3313831.3376523
- [30] Ryo Suzuki, Ryosuke Nakayama, Dan Liu, Yasuaki Kakehi, Mark D. Gross, and Daniel Leithinger. 2020. LiftTiles: Constructive building blocks for prototyping room-scale shape-changing interfaces. *Proceedings of the International Conference on Tangible, Embedded, and Embodied Interaction* (2 2020), 143–151. doi:10.1145/3374920.3374941
- [31] Ryo Suzuki, Clement Zheng, Yasuaki Kakehi, Tom Yeh, Ellen Yi Luen Do, Mark D. Gross, and Daniel Leithinger. 2019. ShapeBots: Shape-changing swarm robots. *Proceedings of the Annual ACM Symposium on User Interface Software and Technology* (10 2019), 493–505. doi:10.1145/3332165.3347911
- [32] Susumu Tachi, Yasuyuki Inoue, and Fumihiro Kato. 2020. TELESAR VI: Telexistence Surrogate Anthropomorphic Robot VI. *International Journal of Humanoid Robotics* (2020), Vol. 17, No. 5, 1–33. https://tachilab.org/en/projects/telesar_vi.html
- [33] Yan Yixian, Kazuki Takashima, Anthony Tang, Takayuki Tanno, Kazuyuki Fujita, and Yoshifumi Kitamura. 2020. ZoomWalls: Dynamic walls that simulate haptic infrastructure for room-scale VR world. *Proceedings of the Annual ACM Symposium on User Interface Software and Technology* (2020), 223–235. doi:10.1145/3379337.3415859

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